

Getting Started

- Download the **icecream randomized data Excel file** posted in Canvas.
- Add **your name** to cell **A12**, the last cell in the “**Flavor**” column:

	A	B	C	D	E	F
1	Flavor	Sugar Cone	Cup	Waffle Cone	Cake Cone	Pretzel Cone
2	Chocolate	30	34	38	42	42
3	Strawberry	39	27	39	43	30
4	Vanilla	41	37	45	43	25
5	Moose Tracks	34	43	31	27	42
6	Lemon	36	28	45	44	35
7	Neapolitan	41	43	36	26	25
8	Blue Moon	35	26	33	30	39
9	Black Cherry	29	25	43	26	35
10	Orange	45	40	33	30	28
11	Cookie Dough	28	30	39	34	30
12	???'s Favorite	26	36	41	30	37

- Save this as a **CSV** file: **icecream_random.csv**
- Close** Excel.
- Reopen** Excel.
- In Excel, open the **icecream_random.csv** file you generated, using the following selections in the “**Text Import Wizard**”:

- Make sure “**Delimited**” is selected; then click **Next**:

Text Import Wizard - Step 1 of 3

The Text Wizard has determined that your data is Delimited.
If this is correct, choose Next, or choose the data type that best describes your data.

Original data type
Choose the file type that best describes your data:
☒ **Delimited** - Characters such as commas or tabs separate each field.
☐ Fixed width - Fields are aligned in columns with spaces between each field.

Start import at row: 1 File origin: 65001 : Unicode (UTF-8)

☐ My data has headers.

Preview of file C:\Users\jnyhoff\Box\School-UWbox\@fall2025\440\wk11\inclass\11-13\icecream_random_Joe.csv

1 Flavor,Sugar Cone,Cup,Waffle Cone,Cake Cone,Pretzel Cone
 2 Chocolate,30,34,38,42,42
 3 Strawberry,39,27,39,43,30
 4 Vanilla,41,37,45,43,25
 5 Moose Tracks,34,43,31,27,42
 6 Lemon,36,28,45,44,35

Cancel < Back **Next >** Finish

- Deselect** **Tab** and **select** **Comma** as the Delimiters; then click **Finish**:

Text Import Wizard - Step 2 of 3

This screen lets you set the delimiters your data contains. You can see how your text is affected in the preview below.

Delimiters
☐ Tab
☐ Semicolon
☒ **Comma**
☐ Space
☐ Other:

☐ Treat consecutive delimiters as one
 Text qualifier: " " " " " " " "

Data preview

Flavor	Sugar Cone	Cup	Waffle Cone	Cake Cone	Pretzel Cone
Chocolate	30	34	38	42	42
Strawberry	39	27	39	43	30
Vanilla	41	37	45	43	25
Moose Tracks	34	43	31	27	42
Lemon	36	28	45	44	35

Cancel < Back **Next >** **Finish**

- Your **icecream_random.csv** file in Excel should look something like this:

	A	B	C	D	E	F
1	Flavor	Sugar Cone	Cup	Waffle Cone	Cake Cone	Pretzel Cone
2	Chocolate	30	34	38	42	42
3	Strawberry	39	27	39	43	30
4	Vanilla	41	37	45	43	25
5	Moose Tracks	34	43	31	27	42
6	Lemon	36	28	45	44	35
7	Neapolitan	41	43	36	26	25
8	Blue Moon	35	26	33	30	39
9	Black Cherry	29	25	43	26	35
10	Orange	45	40	33	30	28
11	Cookie Dough	28	30	39	34	30
12	???’s Favorite	26	36	41	30	37

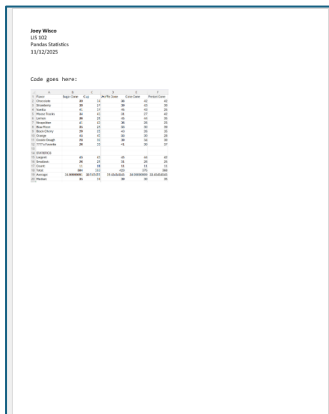
- Because this data is generated **randomly**, your actual data will be different than the above.
- Resave a copy of this CSV file as: **icecream_random_withstats.csv**

Calculating Stats in Excel

- Using **Excel functions**, calculate the following summary data:

	A	B	C	D	E	F
1	Flavor	Sugar Cone	Cup	Waffle Cone	Cake Cone	Pretzel Cone
2	Chocolate	30	34	38	42	42
3	Strawberry	39	27	39	43	30
4	Vanilla	41	37	45	43	25
5	Moose Tracks	34	43	31	27	42
6	Lemon	36	28	45	44	35
7	Neapolitan	41	43	36	26	25
8	Blue Moon	35	26	33	30	39
9	Black Cherry	29	25	43	26	35
10	Orange	45	40	33	30	28
11	Cookie Dough	28	30	39	34	30
12	???’s Favorite	26	36	41	30	37
13						
14	STATISTICS	Use Excel Functions to calculate all of the statistics below:				
15	Largest:	45	43	45	44	42
16	Smallest:	26	25	31	26	25
17	Count:	11	11	11	11	11
18	Total:	384	369	423	375	368
19	Average:	34.90909091	33.54545455	38.45454545	34.09090909	33.45454545
20	Median:	35	34	39	30	35

- Because your ice cream data is different, your summary statistics will be different from the above
- Also, type in the labels you see in cells A14 through A20.
- Copy and Paste a screenshot** of this data into the Microsoft Word template that we have been using in this course.



Calculating Stats in Pandas

- Using code very similar to what we wrote in class for the lecture demo, use Pandas functions/methods with your `icecream_random.csv` file to perform these operations for Sugar Cone, across all 11 flavors:

- total
- largest
- smallest
- count
- median
- average

- Your code output should look like the following:

```
>>> %Run PandasStats.py
Joe Wisco's code...

Sugar Cone Sales:
0      30
1      39
2      41
3      34
4      36
5      41
6      35
7      29
8      45
9      28
10     26
Name: Sugar Cone, dtype: int64
Total Sugar Cone Sales:
384
Largest Sugar Cone Sales:
45
Largest Sugar Cone Sales:
26
Number of Sugar Cone Sales Data Points:
11
Median Sugar Cone Sales for All Flavors:
35.0
Average Sugar Cone Sales for All Flavors:
34.90909090909091
```

- Compare your results to what you calculated in Excel to make sure they match:

```
>>> %Run PandasStats.py
Joe Wisco's code...

Sugar Cone Sales:
0      30
1      39
2      41
3      34
4      36
5      41
6      35
7      29
8      45
9      28
10     26
Name: Sugar Cone, dtype: int64
Total Sugar Cone Sales:
384
Largest Sugar Cone Sales:
45
Largest Sugar Cone Sales:
26
Number of Sugar Cone Sales Data Points:
11
Median Sugar Cone Sales for All Flavors:
35.0
Average Sugar Cone Sales for All Flavors:
34.90909090909091
```



	Min	Q1	Q2	Q3	Max
10 Statistics					
10 Largest	45	41	41	44	42
10 Smallest	26	25	31	26	25
10 Count	11	11	11	11	11
10 Total	384	389	423	375	360
10 Average	34.90909091	35.36363636	38.45454545	34.09090909	32.72727273
10 Median	35	34	35	36	35

- Copy and Paste all of your code and all of your console output for the Sugar Cone sales into the Microsoft Word Template.

Calculating Stats for the Other Containers in Pandas

- Add code to your program to perform the same results for **Cup** sales across all 11 flavors.
- Add code to your program to perform the same results for **Waffle Cone** sales across all 11 flavors.
- Add code to your program to perform the same results for **Cake Cone** sales across all 11 flavors.
- Add code to your program to perform the same results for **Pretzel Cone** sales across all 11 flavors.
- Copy all of your code and all of your console output for these flavors into the Microsoft Word template.

Submission

• PDF

- Save your Microsoft Word template document as a **PDF**.
- Submit this PDF file in the link provided in Canvas.

• CSV

- Submit also your two CSV files:
 - `icecream_random.csv`
 - `icecream_random_withstats.csv`